Conditioning and Preconditioned Iterative Solvers for Sparse Systems from the Semi-Implicit Linearized Shallow Water Equations

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Abstract

This work proposes a controlled numerical study of sparse linear systems arising from a semi-implicit finite-difference discretization of the two-dimensional linearized shallow water equations. Rather than developing a full oceanographic simulation, the equations are used as a model problem to investigate how spatial resolution and variations in water depth affect the conditioning of the discrete system and the convergence of iterative solvers.

The experiments will consider a constant-depth configuration and spatially heterogeneous depth profiles with controlled coefficient contrast. The Conjugate Gradient method without preconditioning will be compared with Jacobi-preconditioned and incomplete-Cholesky-preconditioned variants, provided that the resulting matrix is verified to be symmetric positive definite. The evaluation will consider iteration counts, relative residual norms, estimated condition numbers, and computational time. Preconditioner setup costs and the potential reuse of preconditioners across successive time steps will also be investigated.

The methodological contribution is a controlled, cost-aware comparison of these solver strategies across different grid resolutions and coefficient configurations. The study aims to identify the conditions under which preconditioning improves overall computational performance, rather than assessing its effectiveness solely through iteration counts.

Keywords: Shallow water equations; semi-implicit discretization; sparse linear systems; iterative solvers; preconditioning.

1 Introduction

1.1 Application setup

This work investigates the numerical properties of sparse linear systems arising from the semi-implicit discretization of the two-dimensional linearized shallow water equations. The shallow water model is a depth-averaged approximation of fluid dynamics that is applicable when the horizontal length scales are much larger than the vertical depth. It describes the evolution of the free-surface elevation and the depth-averaged horizontal velocity, making it a useful model for studying wave propagation in coastal and oceanographic settings [4, 5].

Let $\Omega \subset \mathbb{R}^2$ be a rectangular computational domain, and let $t \geq 0$ denote time. The variables $\eta(x, y, t)$ and $\mathbf{u}(x, y, t) = (u, v)^\top$ represent, respectively, the free-surface displacement relative to a resting state and the depth-averaged horizontal velocity. The function $H(x, y) > 0$ denotes the undisturbed water depth, g is the gravitational acceleration, and f is a constant Coriolis parameter.

Under a linearization about a resting state, neglecting nonlinear advection and friction, the governing equations are

$$\frac{\partial \eta}{\partial t} + \nabla \cdot (H\mathbf{u}) = 0, \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + g\nabla\eta + f\mathbf{k} \times \mathbf{u} = 0, \tag{2}$$

where $\mathbf{k}$ denotes the vertical unit vector.

The first equation expresses conservation of mass, while the second describes the linearized horizontal momentum balance. The simplified model isolates the gravity-wave dynamics and provides a controlled setting in which the algebraic properties of the numerical discretization can be studied independently of a full nonlinear oceanographic simulation.

The initial experiments will use a periodic rectangular domain. A constant reference depth, $H(x, y) = H_0$, will provide the baseline configuration. Subsequently, spatially heterogeneous depth profiles will be introduced while maintaining $H(x, y) > 0$. Recent studies on shallow water models and structure-preserving numerical formulations provide additional context for these modeling and discretization choices [3, 5].

The spatial discretization will be based on finite differences in conservative flux form. For the temporal discretization, the gravity-wave terms will be treated implicitly, while the Coriolis term will initially be treated explicitly. This semi-implicit strategy couples the unknown free-surface elevation at the new time level to the unknown velocity at that same level.

Writing Δt for the time step and using superscripts n and $n+1$ for consecutive time levels, the discrete equations are

$$\frac{\eta^{n+1} - \eta^n}{\Delta t} + \nabla_h \cdot (H\mathbf{u}^{n+1}) = 0, \tag{3}$$

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} + g\nabla_h \eta^{n+1} + f\mathbf{k} \times \mathbf{u}^n = 0, \tag{4}$$

where ∇_h and $\nabla_h \cdot$ denote the discrete gradient and divergence operators.

Eliminating the new velocity from these equations leads to a linear system for the free-surface elevation:

$$A_h \eta^{n+1} = b_h^n, \tag{5}$$

where

$$A_h = I - g\Delta t^2 \nabla_h \cdot \big(H\nabla_h\big). \tag{6}$$

Here, I is the identity operator and b_h^n is determined by the solution at the previous time level.

The matrix representation of A_h is sparse because the finite-difference operators couple each grid point only to a small number of neighboring points. This sparsity makes the problem suitable for iterative linear solvers.

A compatible flux-form discretization will be selected so that the resulting matrix is symmetric positive definite under the adopted boundary conditions. In particular, if the discrete divergence and gradient operators are negative adjoints and H is represented by a positive diagonal matrix H_h, then

$$A_h = I + g\Delta t^2 G_h^\top H_h G_h, \tag{7}$$

where G_h is the discrete gradient matrix. This representation provides a mathematical basis for investigating the Conjugate Gradient method. Symmetry and positive definiteness will be verified for the implemented system before this method is used.

The classical semi-implicit shallow water formulation of Casulli [4] and the conjugate-gradient approach studied by Bonaventura and Rosatti [2] provide the principal methodological foundations for this part of the work.

The experiments will investigate two main factors: spatial resolution and coefficient heterogeneity. First, the grid will be progressively refined while

the time step and the remaining parameters are held fixed. This experiment will examine how the dimension and spectral properties of A_h change with spatial resolution.

Second, the depth coefficient will be varied while the grid and time step remain fixed. The heterogeneity will be quantified by the contrast ratio

$$\chi = \frac{H_{\max}}{H_{\min}}, \tag{8}$$

where $H_{\min} > 0$. This experiment will investigate whether changes in coefficient contrast are associated with changes in the conditioning of the matrix and the convergence behavior of the iterative solvers.

The principal methods to be compared are the unpreconditioned Conjugate Gradient method, Jacobi-preconditioned Conjugate Gradient, and incomplete-Cholesky-preconditioned Conjugate Gradient. The choice of these methods is motivated by the literature on iterative methods for sparse linear systems and preconditioning techniques [7, 1]. More recent research also investigates advanced preconditioning strategies for hyperbolic PDE systems, including numerical experiments involving shallow water equations [6].

The evaluation will record the iteration count, relative residual norm, estimated condition number, and computational time. Preconditioner construction time and solution time will be reported separately. Since the coefficient matrix remains unchanged across time steps when the grid, depth profile, and time step are fixed, the study will also examine the effect of reusing a preconditioner across repeated linear solves.

This cost-aware comparison constitutes the principal methodological extension of the work: rather than evaluating preconditioners only by their iteration counts, it will examine when the reduction in solution cost compensates for the additional setup cost. The implementation will use consistent stopping tolerances and initial guesses to ensure meaningful comparisons.

The main research question is therefore:

> How do spatial refinement and depth-coefficient heterogeneity affect the conditioning and iterative convergence of sparse linear systems arising from the semi-implicit shallow water equations, and under what conditions does incomplete Cholesky preconditioning provide a computational advantage over unpreconditioned and Jacobi-preconditioned Conjugate Gradient?

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